

## 29. Programmable Pulse Generator (PPG)

The programmable pulse generator (PPG) generates pulse outputs by using the 16-bit timer pulse unit (TPU) and the multi-function timer pulse unit (MTU) as a time base.

This MCU has two PPG units, each of which controls up to 16 pulse output pins. The pulse outputs from the PPGs are divided into 4-bit groups that can operate all simultaneously and independently.

### 29.1 Overview

Table 29.1 lists the specifications of the PPG and Table 29.2 lists PPG functions.

Figure 29.1 and Figure 29.2 show block diagrams of the PPGs.

**Table 29.1 Specifications of PPG**

| Item                                | Specifications                                                                                                                                                                                                                                      |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of output bits               | Up to 32 bits*1                                                                                                                                                                                                                                     |
| Pulse output                        | <ul style="list-style-type: none"> <li>Two units, each capable of output through four pin groups</li> <li>Output trigger signals are selectable.</li> <li>Non-overlapping operation is possible.</li> <li>Inverted output is selectable.</li> </ul> |
| Output data transfer                | Can operate together with the DTC and DMAC (when TPU and MTU interrupts are in use)                                                                                                                                                                 |
| Power consumption reducing function | Module-stop state can be set for each unit.                                                                                                                                                                                                         |

Note 1. When setting PPG output trigger in MTU, make settings so that PCLKA run at the same frequency as PCLKB.

**Table 29.2 List of PPG Functions**

| Item                            |                                             |               | PPG0                       | PPG1                       |
|---------------------------------|---------------------------------------------|---------------|----------------------------|----------------------------|
| PPG output trigger              | MTU channels 0 to 3 (MTU0 to MTU3)*1        | Compare match | ✓                          | ✓                          |
|                                 |                                             | Input capture | ✓                          | ✓                          |
|                                 | TPU (unit 0) channels 0 to 3 (TPU0 to TPU3) | Compare match | —                          | ✓                          |
|                                 |                                             | Input capture | —                          | ✓                          |
| Non-overlapping operation       |                                             |               | ✓                          | ✓                          |
| Output data transfer            | DTC                                         |               | ✓                          | ✓                          |
|                                 | DMAC                                        |               | ✓                          | ✓                          |
| Selecting inverted output       |                                             |               | ✓                          | ✓                          |
| Setting the module-stop state*2 |                                             |               | The MSTPA11 bit in MSTPCRA | The MSTPA10 bit in MSTPCRA |

✓: Possible

—: Not possible

Note 1. When setting PPG output trigger in MTU, make settings so that PCLKA run at the same frequency as PCLKB.

Note 2. For details, refer to section 11, Low Power Consumption.

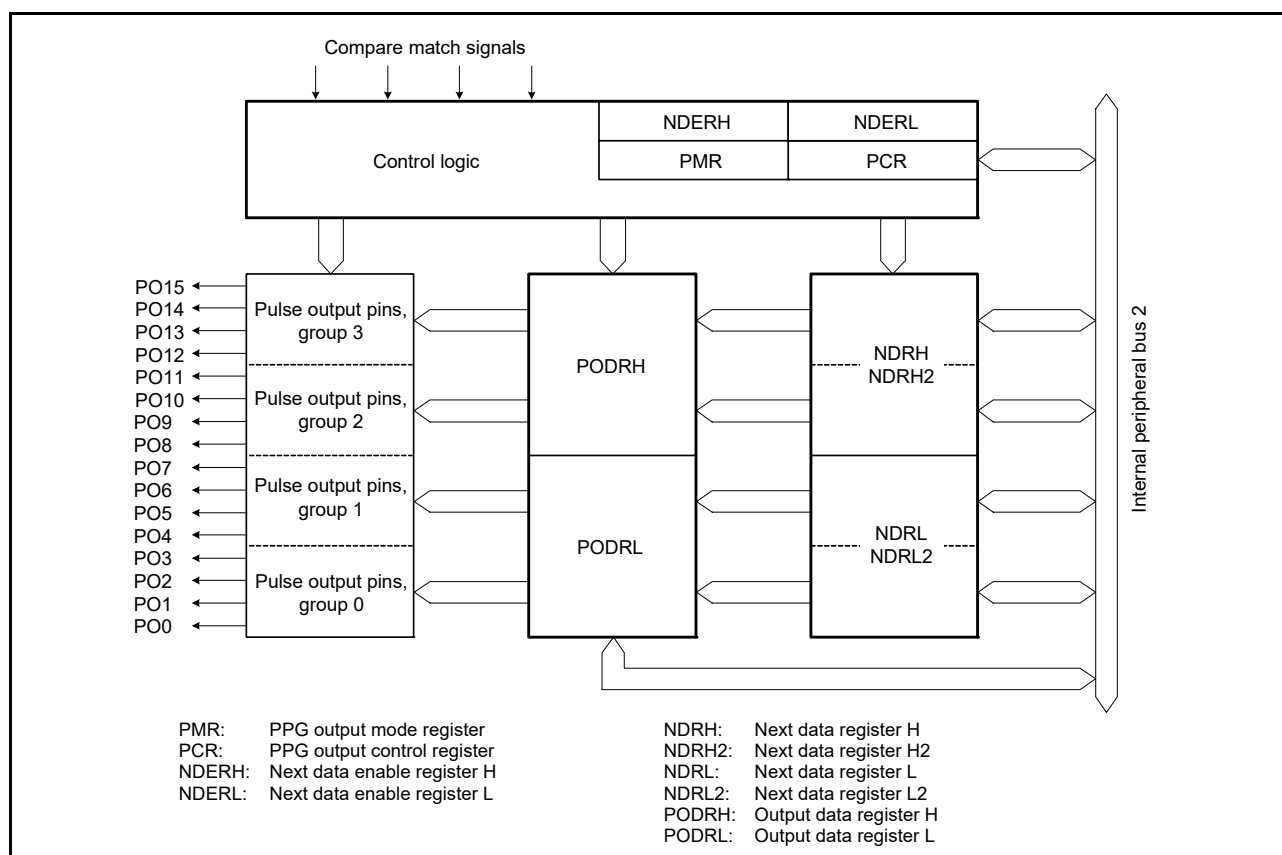


Figure 29.1 Block Diagram of PPG0

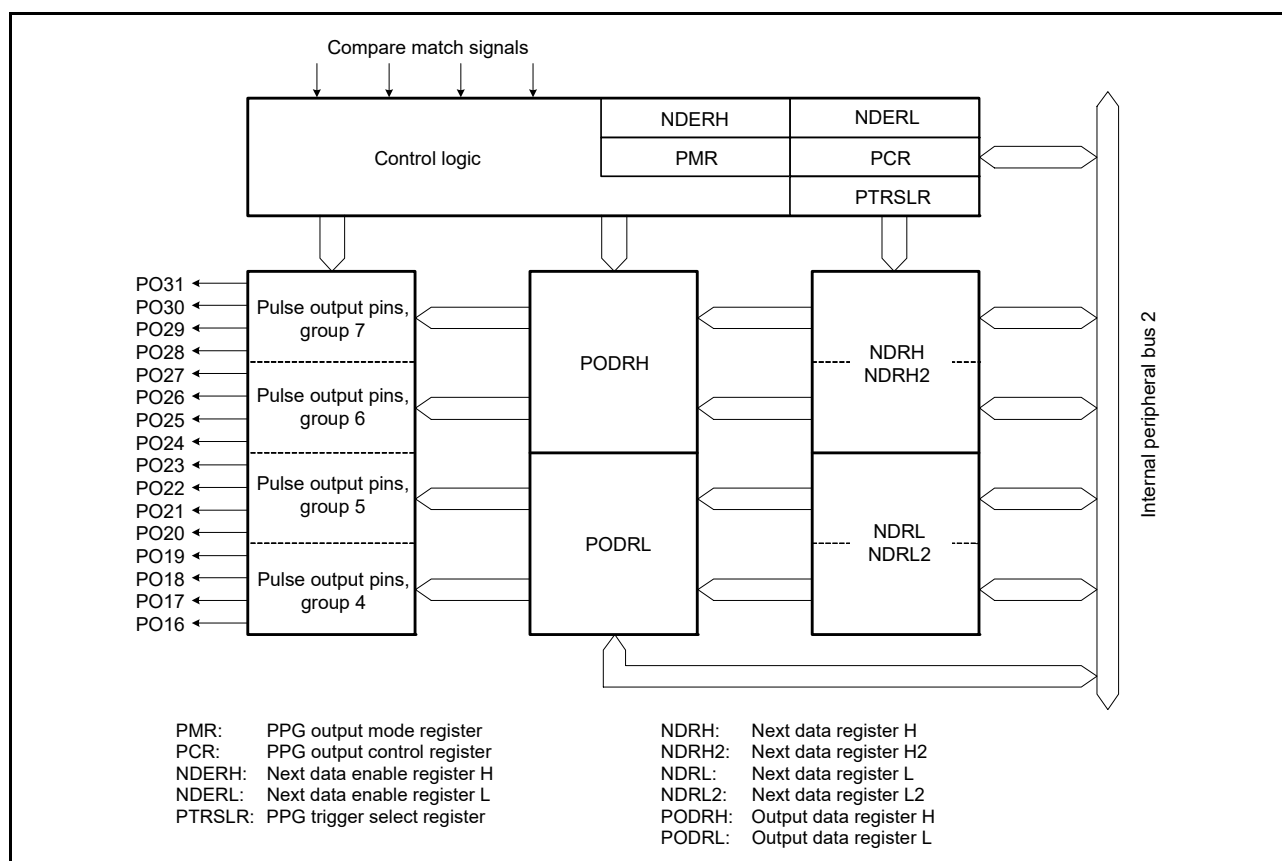


Figure 29.2 Block Diagram of PPG1

Table 29.3 lists the pin configuration of the PPG.

**Table 29.3 Pin Configuration of PPG**

| Unit | Pin Name | I/O    | Function             |
|------|----------|--------|----------------------|
| PPG0 | PO0      | Output | Group 0 pulse output |
|      | PO1      | Output |                      |
|      | PO2      | Output |                      |
|      | PO3      | Output |                      |
|      | PO4      | Output | Group 1 pulse output |
|      | PO5      | Output |                      |
|      | PO6      | Output |                      |
|      | PO7      | Output |                      |
|      | PO8      | Output | Group 2 pulse output |
|      | PO9      | Output |                      |
|      | PO10     | Output |                      |
|      | PO11     | Output |                      |
|      | PO12     | Output | Group 3 pulse output |
|      | PO13     | Output |                      |
|      | PO14     | Output |                      |
|      | PO15     | Output |                      |
| PPG1 | PO16     | Output | Group 4 pulse output |
|      | PO17     | Output |                      |
|      | PO18     | Output |                      |
|      | PO19     | Output |                      |
|      | PO20     | Output | Group 5 pulse output |
|      | PO21     | Output |                      |
|      | PO22     | Output |                      |
|      | PO23     | Output |                      |
|      | PO24     | Output | Group 6 pulse output |
|      | PO25     | Output |                      |
|      | PO26     | Output |                      |
|      | PO27     | Output |                      |
|      | PO28     | Output | Group 7 pulse output |
|      | PO29     | Output |                      |
|      | PO30     | Output |                      |
|      | PO31     | Output |                      |

## 29.2 Register Descriptions

### 29.2.1 PPG Trigger Select Register (PTRSLR)

Address(es): 0008 81F0h

| b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0    |
|----|----|----|----|----|----|----|-------|
| —  | —  | —  | —  | —  | —  | —  | PTRSL |
| 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1     |

Value after reset:

- PPG1.PTRSLR

| Bit      | Symbol | Bit Name           | Description                                                                                                                                  | R/W |
|----------|--------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | PTRSL  | PPG Trigger Select | 0: Selects the set of MTU0 to MTU3 as the trigger channels for PPG1.<br>1: Selects the set of TPU0 to TPU3 as the trigger channels for PPG1. | R/W |
| b7 to b1 | —      | Reserved           | These bits are read as 0. The write value should be 0.                                                                                       | R/W |

#### PTRSL Bit (PPG Trigger Select)

This bit selects either MTU0 to MTU3 or TPU0 to TPU3 as a set of trigger channels for PPG1.

When this bit is set to 0, MTU0 to MTU3 are selected as a set of trigger channels for PPG1. When it is set to 1, TPU0 to TPU3 are selected as a set of trigger channels for PPG1.

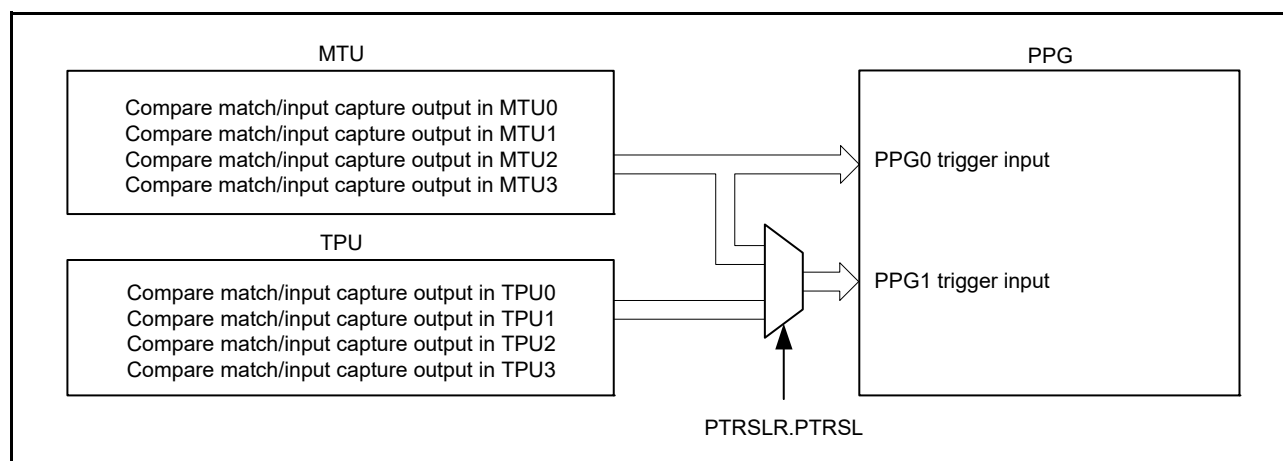


Figure 29.3 Block Diagram of PPG Trigger Selection

### 29.2.2 Next Data Enable Register H (NDERH), Next Data Enable Register L (NDERL)

Address(es): 0008 81E8h

|                    | b7     | b6     | b5     | b4     | b3     | b2     | b1    | b0    |
|--------------------|--------|--------|--------|--------|--------|--------|-------|-------|
| PPG0.NDERH         | NDER15 | NDER14 | NDER13 | NDER12 | NDER11 | NDER10 | NDER9 | NDER8 |
| Value after reset: | 0      | 0      | 0      | 0      | 0      | 0      | 0     | 0     |

Address(es): 0008 81E9h

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| PPG0.NDERL         | NDER7 | NDER6 | NDER5 | NDER4 | NDER3 | NDER2 | NDER1 | NDER0 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

- PPG0.NDERH

| Bit | Symbol | Bit Name                  | Description                                                   | R/W |
|-----|--------|---------------------------|---------------------------------------------------------------|-----|
| b0  | NDER8  | Next Data Transfer Enable | 0: Data transfer is disabled.<br>1: Data transfer is enabled. | R/W |
| b1  | NDER9  | Next Data Transfer Enable |                                                               | R/W |
| b2  | NDER10 | Next Data Transfer Enable |                                                               | R/W |
| b3  | NDER11 | Next Data Transfer Enable |                                                               | R/W |
| b4  | NDER12 | Next Data Transfer Enable |                                                               | R/W |
| b5  | NDER13 | Next Data Transfer Enable |                                                               | R/W |
| b6  | NDER14 | Next Data Transfer Enable |                                                               | R/W |
| b7  | NDER15 | Next Data Transfer Enable |                                                               | R/W |

PPG0.NDERH selects the pins (PO15 to PO8) for outputs of pulse from the PPG on a bit-by-bit basis.

#### NDERi Bits (Next Data Transfer Enable) (i = 15 to 8)

For NDERi bits with the setting 1, the trigger specified by PPG0.PCR leads to the transfer of the values from the corresponding bits of the PPG0.NDRH or PPG0.NDRH2 register to those of the PPG0.PODRH register. For NDERi bits with the setting 0, the values are not transferred from the corresponding bits of the PPG0.NDRH or PPG0.NDRH2 register to those of the PPG0.PODRH register.

- PPG0.NDERL

| Bit | Symbol | Bit Name                  | Description                                                   | R/W |
|-----|--------|---------------------------|---------------------------------------------------------------|-----|
| b0  | NDER0  | Next Data Transfer Enable | 0: Data transfer is disabled.<br>1: Data transfer is enabled. | R/W |
| b1  | NDER1  | Next Data Transfer Enable |                                                               | R/W |
| b2  | NDER2  | Next Data Transfer Enable |                                                               | R/W |
| b3  | NDER3  | Next Data Transfer Enable |                                                               | R/W |
| b4  | NDER4  | Next Data Transfer Enable |                                                               | R/W |
| b5  | NDER5  | Next Data Transfer Enable |                                                               | R/W |
| b6  | NDER6  | Next Data Transfer Enable |                                                               | R/W |
| b7  | NDER7  | Next Data Transfer Enable |                                                               | R/W |

PPG0.NDERL selects the pins (PO7 to PO0) for outputs of pulse from the PPG on a bit-by-bit basis.

**NDERi Bits (Next Data Transfer Enable) (i = 7 to 0)**

For NDERi bits with the setting 1, the trigger specified by PPG0.PCR leads to the transfer of the values from the corresponding bits of the PPG0.NDRL or PPG0.NDRL2 register to those of the PPG0.PODRL register. For NDERi bits with the setting 0, the values are not transferred from the corresponding bits of the PPG0.NDRL or PPG0.NDRL2 register to those of the PPG0.PODRL register.

Address(es): 0008 81F8h

|                    | b7      | b6      | b5      | b4      | b3      | b2      | b1      | b0      |
|--------------------|---------|---------|---------|---------|---------|---------|---------|---------|
| PPG1.NDERH         | NDER 31 | NDER 30 | NDER 29 | NDER 28 | NDER 27 | NDER 26 | NDER 25 | NDER 24 |
| Value after reset: | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       |

Address(es): 0008 81F9h

|                    | b7      | b6      | b5      | b4      | b3      | b2      | b1      | b0      |
|--------------------|---------|---------|---------|---------|---------|---------|---------|---------|
| PPG1.NDERL         | NDER 23 | NDER 22 | NDER 21 | NDER 20 | NDER 19 | NDER 18 | NDER 17 | NDER 16 |
| Value after reset: | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       |

- PPG1.NDERH

| Bit | Symbol  | Bit Name                  | Description                   | R/W |
|-----|---------|---------------------------|-------------------------------|-----|
| b0  | NDER 24 | Next Data Transfer Enable | 0: Data transfer is disabled. | R/W |
| b1  | NDER 25 | Next Data Transfer Enable | 1: Data transfer is enabled.  | R/W |
| b2  | NDER 26 | Next Data Transfer Enable |                               | R/W |
| b3  | NDER 27 | Next Data Transfer Enable |                               | R/W |
| b4  | NDER 28 | Next Data Transfer Enable |                               | R/W |
| b5  | NDER 29 | Next Data Transfer Enable |                               | R/W |
| b6  | NDER 30 | Next Data Transfer Enable |                               | R/W |
| b7  | NDER 31 | Next Data Transfer Enable |                               | R/W |

PPG1.NDERH selects the pins (PO31 to PO24) for outputs of pulse from the PPG on a bit-by-bit basis.

**NDERi Bits (Next Data Transfer Enable) (i = 31 to 24)**

For NDERi bits with the setting 1, the trigger specified by PPG1.PCR leads to the transfer of the values from the corresponding bits of the PPG1.NDRH or PPG1.NDRH2 register to those of the PPG1.PODRH register. For NDERi bits with the setting 0, the values are not transferred from the corresponding bits of the PPG1.NDRH or PPG1.NDRH2 register to those of the PPG1.PODRH register.

- PPG1.NDERL

| Bit | Symbol  | Bit Name                  | Description                   | R/W |
|-----|---------|---------------------------|-------------------------------|-----|
| b0  | NDER 16 | Next Data Transfer Enable | 0: Data transfer is disabled. | R/W |
| b1  | NDER 17 | Next Data Transfer Enable | 1: Data transfer is enabled.  | R/W |
| b2  | NDER 18 | Next Data Transfer Enable |                               | R/W |
| b3  | NDER 19 | Next Data Transfer Enable |                               | R/W |
| b4  | NDER 20 | Next Data Transfer Enable |                               | R/W |
| b5  | NDER 21 | Next Data Transfer Enable |                               | R/W |
| b6  | NDER 22 | Next Data Transfer Enable |                               | R/W |
| b7  | NDER 23 | Next Data Transfer Enable |                               | R/W |

PPG1.NDERL selects the pins (PO23 to PO16) for outputs of pulse from the PPG on a bit-by-bit basis.

**NDERi Bits (Next Data Transfer Enable) (i = 23 to 16)**

For NDERi bits with the setting 1, the trigger specified by PPG1.PCR leads to the transfer of the values from the corresponding bits of the PPG1.NDRL or PPG1.NDRL2 register to those of the PPG1.PODRL register. For NDERi bits with the setting 0, the values are not transferred from the corresponding bits of the PPG1.NDRL or PPG1.NDRL2 register to those of the PPG1.PODRL register.

### 29.2.3 Output Data Register H (PODRH), Output Data Register L (PODRL)

Address(es): 0008 81EAh

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1   | b0   |
|--------------------|-------|-------|-------|-------|-------|-------|------|------|
| PPG0.PODRH         | POD15 | POD14 | POD13 | POD12 | POD11 | POD10 | POD9 | POD8 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |

Address(es): 0008 81EBh

|                    | b7   | b6   | b5   | b4   | b3   | b2   | b1   | b0   |
|--------------------|------|------|------|------|------|------|------|------|
| PPG0.PODRL         | POD7 | POD6 | POD5 | POD4 | POD3 | POD2 | POD1 | POD0 |
| Value after reset: | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |

- PPG0.PODRH

| Bit | Symbol | Bit Name             | Description                                 | R/W |
|-----|--------|----------------------|---------------------------------------------|-----|
| b0  | POD8   | Output Data Register | 0: The low level is output on the POi pin.  | R/W |
| b1  | POD9   | Output Data Register | 1: The high level is output on the POi pin. | R/W |
| b2  | POD10  | Output Data Register | (i = 15 to 8)                               | R/W |
| b3  | POD11  | Output Data Register |                                             | R/W |
| b4  | POD12  | Output Data Register |                                             | R/W |
| b5  | POD13  | Output Data Register |                                             | R/W |
| b6  | POD14  | Output Data Register |                                             | R/W |
| b7  | POD15  | Output Data Register |                                             | R/W |

PPG0.PODRH stores pulse output values. For bits corresponding to pins that have been set for pulse output by PPG0.NDERH, the output trigger transfers the values in PPG0.NDRH or PPG0.NDRH2 to this register.

#### PODi Bit (Output Data Register) (i = 15 to 8)

When an output trigger is generated during PPG operation, the values of bits for which data transfer is enabled in the PPG0.NDERH register are transferred from the PPG0.NDRH or PPG0.NDRH2 register to this register. Writing from the CPU is impossible while any of the NDERi (i = 15 to 8) bits in PPG0.NDERH is 1. The initial output level for the pulse signal can be set when the value in the PPG0.NDERH register is 00h.

- PPG0.PODRL

| Bit | Symbol | Bit Name             | Description                                 | R/W |
|-----|--------|----------------------|---------------------------------------------|-----|
| b0  | POD0   | Output Data Register | 0: The low level is output on the POi pin.  | R/W |
| b1  | POD1   | Output Data Register | 1: The high level is output on the POi pin. | R/W |
| b2  | POD2   | Output Data Register | (i = 7 to 0)                                | R/W |
| b3  | POD3   | Output Data Register |                                             | R/W |
| b4  | POD4   | Output Data Register |                                             | R/W |
| b5  | POD5   | Output Data Register |                                             | R/W |
| b6  | POD6   | Output Data Register |                                             | R/W |
| b7  | POD7   | Output Data Register |                                             | R/W |

PPG0.PODRL stores pulse output values. For bits corresponding to pins that have been set for pulse output by PPG0.NDERL, the output trigger transfers the values in PPG0.NDRL or PPG0.NDRL2 to this register.

**PODi Bit (Output Data Register) (i = 7 to 0)**

When an output trigger is generated during PPG operation, the values of bits for which data transfer is enabled in the PPG0.NDERL register are transferred from the PPG0.NDRL or PPG0.NDRL2 register to this register. Writing from the CPU is impossible while any of the NDERi (i = 7 to 0) bits in PPG0.NDERL is 1. The initial output level for the pulse signal can be set when the value in the PPG0.NDERL register is 00h.

Address(es): 0008 81FAh

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| PPG1.PODRH         | POD31 | POD30 | POD29 | POD28 | POD27 | POD26 | POD25 | POD24 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

Address(es): 0008 81FBh

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| PPG1.PODRL         | POD23 | POD22 | POD21 | POD20 | POD19 | POD18 | POD17 | POD16 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

- PPG1.PODRH

| Bit | Symbol | Bit Name             | Description                                 | R/W |
|-----|--------|----------------------|---------------------------------------------|-----|
| b0  | POD24  | Output Data Register | 0: The low level is output on the POi pin.  | R/W |
| b1  | POD25  | Output Data Register | 1: The high level is output on the POi pin. | R/W |
| b2  | POD26  | Output Data Register | (i = 31 to 24)                              | R/W |
| b3  | POD27  | Output Data Register |                                             | R/W |
| b4  | POD28  | Output Data Register |                                             | R/W |
| b5  | POD29  | Output Data Register |                                             | R/W |
| b6  | POD30  | Output Data Register |                                             | R/W |
| b7  | POD31  | Output Data Register |                                             | R/W |

PPG1.PODRH stores pulse output values. For bits corresponding to pins that have been set for pulse output by PPG1.NDERH, the output trigger transfers the values in PPG1.NDRH or PPG1.NDRH2 to this register.

**PODi Bit (Output Data Register) (i = 31 to 24)**

When an output trigger is generated during PPG operation, the values of bits for which data transfer is enabled in the PPG1.NDERH register are transferred from the PPG1.NDRH or PPG1.NDRH2 register to this register. Writing from the CPU is impossible while any of the NDERi (i = 31 to 24) bits in PPG1.NDERH is 1. The initial output level for the pulse signal can be set when the value in the PPG1.NDERH register is 00h.

- PPG1.PODRL

| Bit | Symbol | Bit Name             | Description                                 | R/W |
|-----|--------|----------------------|---------------------------------------------|-----|
| b0  | POD16  | Output Data Register | 0: The low level is output on the POi pin.  | R/W |
| b1  | POD17  | Output Data Register | 1: The high level is output on the POi pin. | R/W |
| b2  | POD18  | Output Data Register | (i = 23 to 16)                              | R/W |
| b3  | POD19  | Output Data Register |                                             | R/W |
| b4  | POD20  | Output Data Register |                                             | R/W |
| b5  | POD21  | Output Data Register |                                             | R/W |
| b6  | POD22  | Output Data Register |                                             | R/W |
| b7  | POD23  | Output Data Register |                                             | R/W |

PPG1.PODRL stores pulse output values. For bits corresponding to pins that have been set for pulse output by PPG1.NDERL, the output trigger transfers the values in PPG1.NDRL or PPG1.NDRL2 to this register.

**PODi Bit (Output Data Register) (i = 23 to 16)**

When an output trigger is generated during PPG operation, the values of bits for which data transfer is enabled in the PPG1.NDERL register are transferred from the PPG1.NDRL or PPG1.NDRL2 register to this register. Writing from the CPU is impossible while any of the NDERi (i = 23 to 16) bits in PPG1.NDERL is 1. The initial output level for the pulse signal can be set when the value in the PPG1.NDERL register is 00h.

## 29.2.4 Next Data Register H (NDRH), Next Data Register L (NDRL), Next Data Register H2 (NDRH2), Next Data Register L2 (NDRL2)

Address(es): 0008 81ECh

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1   | b0   |
|--------------------|-------|-------|-------|-------|-------|-------|------|------|
| PPG0.NDRH          | NDR15 | NDR14 | NDR13 | NDR12 | NDR11 | NDR10 | NDR9 | NDR8 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |

Address(es): 0008 81EDh

|                    | b7   | b6   | b5   | b4   | b3   | b2   | b1   | b0   |
|--------------------|------|------|------|------|------|------|------|------|
| PPG0.NDRL          | NDR7 | NDR6 | NDR5 | NDR4 | NDR3 | NDR2 | NDR1 | NDR0 |
| Value after reset: | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |

Address(es): 0008 81EEh

|                    | b7 | b6 | b5 | b4 | b3    | b2    | b1   | b0   |
|--------------------|----|----|----|----|-------|-------|------|------|
| PPG0.NDRH2         | —  | —  | —  | —  | NDR11 | NDR10 | NDR9 | NDR8 |
| Value after reset: | 0  | 0  | 0  | 0  | 0     | 0     | 0    | 0    |

Address(es): 0008 81EFh

|                    | b7 | b6 | b5 | b4 | b3   | b2   | b1   | b0   |
|--------------------|----|----|----|----|------|------|------|------|
| PPG0.NDRL2         | —  | —  | —  | —  | NDR3 | NDR2 | NDR1 | NDR0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0    | 0    | 0    | 0    |

- PPG0.NDRH

The PPG0.NDRH register holds the next pulse output value to the one stored in the PODRH register. The assignment of bits in the PPG0.NDRH register depends on whether pulse output groups are set as having the same output trigger or different output triggers.

(1) When pulse output groups 2 and 3 have the same output trigger

If pulse output groups 2 and 3 have the same output trigger, all eight bits are mapped to the NDRH register and can be accessed in a single operation.

| Bit | Symbol | Bit Name           | Description                                                                                                             | R/W |
|-----|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | NDR8   | Next Data Register | The output trigger specified by PPG0.PCR transfers the values in this register to the corresponding bits in PPG0.PODRH. | R/W |
| b1  | NDR9   | Next Data Register |                                                                                                                         | R/W |
| b2  | NDR10  | Next Data Register |                                                                                                                         | R/W |
| b3  | NDR11  | Next Data Register |                                                                                                                         | R/W |
| b4  | NDR12  | Next Data Register |                                                                                                                         | R/W |
| b5  | NDR13  | Next Data Register |                                                                                                                         | R/W |
| b6  | NDR14  | Next Data Register |                                                                                                                         | R/W |
| b7  | NDR15  | Next Data Register |                                                                                                                         | R/W |

## (2) When pulse output groups 2 and 3 have different output triggers

If different output triggers are selected for pulse output groups 2 and 3, the pulse output group 3 are mapped to the upper 4 bits.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |
| b4       | NDR12  | Next Data Register | The output trigger specified by PPG0.PCR transfers the values in this register to the corresponding bits in PPG0.PODRH. | R/W |
| b5       | NDR13  | Next Data Register |                                                                                                                         | R/W |
| b6       | NDR14  | Next Data Register |                                                                                                                         | R/W |
| b7       | NDR15  | Next Data Register |                                                                                                                         | R/W |

- PPG0.NDRH2

If different output triggers are selected for pulse output groups 2 and 3, the pulse output group 2 are mapped to the lower 4 bits of the PPG0.NDRH2 register.

If the same output trigger is selected for pulse output groups 2 and 3, the PPG0.NDRH2 register is not used. With such settings, the value read is FFh and writing is prohibited.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | NDR8   | Next Data Register | The output trigger specified by PPG0.PCR transfers the values in this register to the corresponding bits in PPG0.PODRH. | R/W |
| b1       | NDR9   | Next Data Register |                                                                                                                         | R/W |
| b2       | NDR10  | Next Data Register |                                                                                                                         | R/W |
| b3       | NDR11  | Next Data Register |                                                                                                                         | R/W |
| b7 to b4 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |

- PPG0.NDRL

The PPG0.NDRL register holds the next pulse output value to the one stored in the PODRL register. The assignment of bits in the PPG0.NDRL register depends on whether pulse output groups are set as having the same output trigger or different output triggers.

## (1) When pulse output groups 0 and 1 have the same output trigger

If pulse output groups 0 and 1 have the same output trigger, all eight bits are mapped to the NDRL register and can be accessed in a single operation.

| Bit | Symbol | Bit Name           | Description                                                                                                             | R/W |
|-----|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | NDR0   | Next Data Register | The output trigger specified by PPG0.PCR transfers the values in this register to the corresponding bits in PPG0.PODRL. | R/W |
| b1  | NDR1   | Next Data Register |                                                                                                                         | R/W |
| b2  | NDR2   | Next Data Register |                                                                                                                         | R/W |
| b3  | NDR3   | Next Data Register |                                                                                                                         | R/W |
| b4  | NDR4   | Next Data Register |                                                                                                                         | R/W |
| b5  | NDR5   | Next Data Register |                                                                                                                         | R/W |
| b6  | NDR6   | Next Data Register |                                                                                                                         | R/W |
| b7  | NDR7   | Next Data Register |                                                                                                                         | R/W |

## (2) When pulse output groups 0 and 1 have different output triggers

If different output triggers are selected for pulse output groups 0 and 1, the pulse output group 1 are mapped to the upper 4 bits.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |
| b4       | NDR4   | Next Data Register | The output trigger specified by PPG0.PCR transfers the values in this register to the corresponding bits in PPG0.PODRL. | R/W |
| b5       | NDR5   | Next Data Register |                                                                                                                         | R/W |
| b6       | NDR6   | Next Data Register |                                                                                                                         | R/W |
| b7       | NDR7   | Next Data Register |                                                                                                                         | R/W |

- PPG0.NDRL2

If different output triggers are selected for pulse output groups 0 and 1, the pulse output group 0 are mapped to the lower 4 bits of the PPG0.NDRL2 register.

If the same output trigger is selected for pulse output groups 0 and 1, the PPG0.NDRL2 register is not used. With such settings, the value read is FFh and writing is prohibited.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | NDR0   | Next Data Register | The output trigger specified by PPG0.PCR transfers the values in this register to the corresponding bits in PPG0.PODRL. | R/W |
| b1       | NDR1   | Next Data Register |                                                                                                                         | R/W |
| b2       | NDR2   | Next Data Register |                                                                                                                         | R/W |
| b3       | NDR3   | Next Data Register |                                                                                                                         | R/W |
| b7 to b4 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |

Address(es): 0008 81FCh

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| PPG1.NDRH          | NDR31 | NDR30 | NDR29 | NDR28 | NDR27 | NDR26 | NDR25 | NDR24 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

Address(es): 0008 81FDh

|                    | b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| PPG1.NDRL          | NDR23 | NDR22 | NDR21 | NDR20 | NDR19 | NDR18 | NDR17 | NDR16 |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

Address(es): 0008 81FEh

|                    | b7 | b6 | b5 | b4 | b3    | b2    | b1    | b0    |
|--------------------|----|----|----|----|-------|-------|-------|-------|
| PPG1.NDRH2         | —  | —  | —  | —  | NDR27 | NDR26 | NDR25 | NDR24 |
| Value after reset: | 0  | 0  | 0  | 0  | 0     | 0     | 0     | 0     |

Address(es): 0008 81FFh

|                    | b7 | b6 | b5 | b4 | b3    | b2    | b1    | b0    |
|--------------------|----|----|----|----|-------|-------|-------|-------|
| PPG1.NDRL2         | —  | —  | —  | —  | NDR19 | NDR18 | NDR17 | NDR16 |
| Value after reset: | 0  | 0  | 0  | 0  | 0     | 0     | 0     | 0     |

- PPG1.NDRH

The PPG1.NDRH register holds the next pulse output value to the one stored in the PODRH register. The assignment of bits in the PPG1.NDRH register depends on whether pulse output groups are set as having the same output trigger or different output triggers.

(1) When pulse output groups 6 and 7 have the same output trigger

If pulse output groups 6 and 7 have the same output trigger, all eight bits are mapped to the NDRH register and can be accessed in a single operation.

| Bit | Symbol | Bit Name           | Description                                                                                                             | R/W |
|-----|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | NDR24  | Next Data Register | The output trigger specified by PPG1.PCR transfers the values in this register to the corresponding bits in PPG1.PODRH. | R/W |
| b1  | NDR25  | Next Data Register |                                                                                                                         | R/W |
| b2  | NDR26  | Next Data Register |                                                                                                                         | R/W |
| b3  | NDR27  | Next Data Register |                                                                                                                         | R/W |
| b4  | NDR28  | Next Data Register |                                                                                                                         | R/W |
| b5  | NDR29  | Next Data Register |                                                                                                                         | R/W |
| b6  | NDR30  | Next Data Register |                                                                                                                         | R/W |
| b7  | NDR31  | Next Data Register |                                                                                                                         | R/W |

## (2) When pulse output groups 6 and 7 have different output triggers

If different output triggers are selected for pulse output groups 6 and 7, the pulse output group 7 are mapped to the upper 4 bits.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |
| b4       | NDR28  | Next Data Register | The output trigger specified by PPG1.PCR transfers the values in this register to the corresponding bits in PPG1.PODRH. | R/W |
| b5       | NDR29  | Next Data Register |                                                                                                                         | R/W |
| b6       | NDR30  | Next Data Register |                                                                                                                         | R/W |
| b7       | NDR31  | Next Data Register |                                                                                                                         | R/W |

- PPG1.NDRH2

If different output triggers are selected for pulse output groups 6 and 7, the pulse output group 6 are mapped to the lower 4 bits of the PPG1.NDRH2 register.

If the same output trigger is selected for pulse output groups 6 and 7, the PPG1.NDRH2 register is not used. With such settings, the value read is FFh and writing is prohibited.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | NDR24  | Next Data Register | The output trigger specified by PPG1.PCR transfers the values in this register to the corresponding bits in PPG1.PODRH. | R/W |
| b1       | NDR25  | Next Data Register |                                                                                                                         | R/W |
| b2       | NDR26  | Next Data Register |                                                                                                                         | R/W |
| b3       | NDR27  | Next Data Register |                                                                                                                         | R/W |
| b7 to b4 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |

- PPG1.NDRL

The PPG1.NDRL register holds the next pulse output value to the one stored in the PODRL register. The assignment of bits in the PPG1.NDRL register depends on whether pulse output groups are set as having the same output trigger or different output triggers.

## (1) When pulse output groups 4 and 5 have the same output trigger

If pulse output groups 4 and 5 have the same output trigger, all eight bits are mapped to the NDRL register and can be accessed in a single operation.

| Bit | Symbol | Bit Name           | Description                                                                                                             | R/W |
|-----|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | NDR16  | Next Data Register | The output trigger specified by PPG1.PCR transfers the values in this register to the corresponding bits in PPG1.PODRL. | R/W |
| b1  | NDR17  | Next Data Register |                                                                                                                         | R/W |
| b2  | NDR18  | Next Data Register |                                                                                                                         | R/W |
| b3  | NDR19  | Next Data Register |                                                                                                                         | R/W |
| b4  | NDR20  | Next Data Register |                                                                                                                         | R/W |
| b5  | NDR21  | Next Data Register |                                                                                                                         | R/W |
| b6  | NDR22  | Next Data Register |                                                                                                                         | R/W |
| b7  | NDR23  | Next Data Register |                                                                                                                         | R/W |

## (2) When pulse output groups 4 and 5 have different output triggers

If different output triggers are selected for pulse output groups 4 and 5, the pulse output group 5 are mapped to the upper 4 bits.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |
| b4       | NDR20  | Next Data Register | The output trigger specified by PPG1.PCR transfers the values in this register to the corresponding bits in PPG1.PODRL. | R/W |
| b5       | NDR21  | Next Data Register |                                                                                                                         | R/W |
| b6       | NDR22  | Next Data Register |                                                                                                                         | R/W |
| b7       | NDR23  | Next Data Register |                                                                                                                         | R/W |

- PPG1.NDRL2

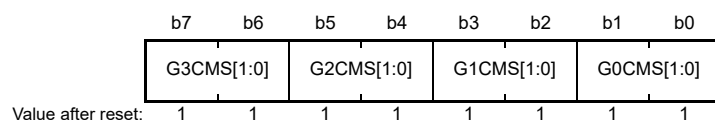
If different output triggers are selected for pulse output groups 4 and 5, the pulse output group 4 are mapped to the lower 4 bits of the PPG1.NDRL2 register.

If the same output trigger is selected for pulse output groups 4 and 5, the PPG1.NDRL2 register is not used. With such settings, the value read is FFh and writing is prohibited.

| Bit      | Symbol | Bit Name           | Description                                                                                                             | R/W |
|----------|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | NDR16  | Next Data Register | The output trigger specified by PPG1.PCR transfers the values in this register to the corresponding bits in PPG1.PODRL. | R/W |
| b1       | NDR17  | Next Data Register |                                                                                                                         | R/W |
| b2       | NDR18  | Next Data Register |                                                                                                                         | R/W |
| b3       | NDR19  | Next Data Register |                                                                                                                         | R/W |
| b7 to b4 | —      | Reserved           | These bits are read as 1. The write value should be 1.                                                                  | R/W |

## 29.2.5 PPG Output Control Register (PCR)

Address(es): PPG0.PCR 0008 81E6h, PPG1.PCR 0008 81F6h



## • PPG0.PCR

| Bit    | Symbol     | Bit Name                     | Description                                                                                                                   | R/W |
|--------|------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----|
| b1, b0 | G0CMS[1:0] | Group 0 Compare Match Select | b1 b0<br>0 0: Compare match in MTU0<br>0 1: Compare match in MTU1<br>1 0: Compare match in MTU2<br>1 1: Compare match in MTU3 | R/W |
| b3, b2 | G1CMS[1:0] | Group 1 Compare Match Select | b3 b2<br>0 0: Compare match in MTU0<br>0 1: Compare match in MTU1<br>1 0: Compare match in MTU2<br>1 1: Compare match in MTU3 | R/W |
| b5, b4 | G2CMS[1:0] | Group 2 Compare Match Select | b5 b4<br>0 0: Compare match in MTU0<br>0 1: Compare match in MTU1<br>1 0: Compare match in MTU2<br>1 1: Compare match in MTU3 | R/W |
| b7, b6 | G3CMS[1:0] | Group 3 Compare Match Select | b7 b6<br>0 0: Compare match in MTU0<br>0 1: Compare match in MTU1<br>1 0: Compare match in MTU2<br>1 1: Compare match in MTU3 | R/W |

## • PPG1.PCR

| Bit    | Symbol     | Bit Name                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W |
|--------|------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b1, b0 | G0CMS[1:0] | Group 4 Compare Match Select | <ul style="list-style-type: none"> <li>When the PTRSL bit in PPG1.PTRSLR is set to 0.</li> </ul> b1 b0<br>0 0: Compare match in MTU0<br>0 1: Compare match in MTU1<br>1 0: Compare match in MTU2<br>1 1: Compare match in MTU3 <ul style="list-style-type: none"> <li>When the PTRSL bit in PPG1.PTRSLR is set to 1.</li> </ul> b1 b0<br>0 0: Compare match in TPU0<br>0 1: Compare match in TPU1<br>1 0: Compare match in TPU2<br>1 1: Compare match in TPU3 | R/W |
| b3, b2 | G1CMS[1:0] | Group 5 Compare Match Select | <ul style="list-style-type: none"> <li>When the PTRSL bit in PPG1.PTRSLR is set to 0.</li> </ul> b3 b2<br>0 0: Compare match in MTU0<br>0 1: Compare match in MTU1<br>1 0: Compare match in MTU2<br>1 1: Compare match in MTU3 <ul style="list-style-type: none"> <li>When the PTRSL bit in PPG1.PTRSLR is set to 1.</li> </ul> b3 b2<br>0 0: Compare match in TPU0<br>0 1: Compare match in TPU1<br>1 0: Compare match in TPU2<br>1 1: Compare match in TPU3 | R/W |

| Bit    | Symbol     | Bit Name                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W |
|--------|------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b5, b4 | G2CMS[1:0] | Group 6 Compare Match Select | <ul style="list-style-type: none"> <li>When the PTRSL bit in PPG1.PTRSLR is set to 0.               <div>b5 b4</div> <div>0 0: Compare match in MTU0</div> <div>0 1: Compare match in MTU1</div> <div>1 0: Compare match in MTU2</div> <div>1 1: Compare match in MTU3</div> </li> <li>When the PTRSL bit in PPG1.PTRSLR is set to 1.               <div>b5 b4</div> <div>0 0: Compare match in TPU0</div> <div>0 1: Compare match in TPU1</div> <div>1 0: Compare match in TPU2</div> <div>1 1: Compare match in TPU3</div> </li> </ul> | R/W |
| b7, b6 | G3CMS[1:0] | Group 7 Compare Match Select | <ul style="list-style-type: none"> <li>When the PTRSL bit in PPG1.PTRSLR is set to 0.               <div>b7 b6</div> <div>0 0: Compare match in MTU0</div> <div>0 1: Compare match in MTU1</div> <div>1 0: Compare match in MTU2</div> <div>1 1: Compare match in MTU3</div> </li> <li>When the PTRSL bit in PPG1.PTRSLR is set to 1.               <div>b7 b6</div> <div>0 0: Compare match in TPU0</div> <div>0 1: Compare match in TPU1</div> <div>1 0: Compare match in TPU2</div> <div>1 1: Compare match in TPU3</div> </li> </ul> | R/W |

PPGn.PCR (n = 0, 1) selects pulse output trigger signals on a group-by-group basis. For details on output trigger selection, refer to section 29.2.6, PPG Output Mode Register (PMR).

## 29.2.6 PPG Output Mode Register (PMR)

Address(es): PPG0.PMR 0008 81E7h, PPG1.PMR 0008 81F7h

| b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|-------|-------|-------|-------|-------|-------|-------|-------|
| G3INV | G2INV | G1INV | G0INV | G3NOV | G2NOV | G1NOV | G0NOV |
| 1     | 1     | 1     | 1     | 0     | 0     | 0     | 0     |

Value after reset:

- PPG0.PMR

| Bit | Symbol | Bit Name                       | Description                                                                                                                                                                                                    | R/W |
|-----|--------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | G0NOV  | Group 0 Non-Overlap            | 0: Normal operation<br>(Output values updated on compare match A in the selected MTUn)<br>1: Non-overlapping operation<br>(Output values updated on compare match A or B in the selected MTUn)<br>(n = 0 to 3) | R/W |
| b1  | G1NOV  | Group 1 Non-Overlap            | 0: Normal operation<br>(Output values updated on compare match A in the selected MTUn)<br>1: Non-overlapping operation<br>(Output values updated on compare match A or B in the selected MTUn)<br>(n = 0 to 3) | R/W |
| b2  | G2NOV  | Group 2 Non-Overlap            | 0: Normal operation<br>(Output values updated on compare match A in the selected MTUn)<br>1: Non-overlapping operation<br>(Output values updated on compare match A or B in the selected MTUn)<br>(n = 0 to 3) | R/W |
| b3  | G3NOV  | Group 3 Non-Overlap            | 0: Normal operation<br>(Output values updated on compare match A in the selected MTUn)<br>1: Non-overlapping operation<br>(Output values updated on compare match A or B in the selected MTUn)<br>(n = 0 to 3) | R/W |
| b4  | G0INV  | Group 0 Output Polarity Change | 0: Inverted output<br>1: Direct output                                                                                                                                                                         | R/W |
| b5  | G1INV  | Group 1 Output Polarity Change | 0: Inverted output<br>1: Direct output                                                                                                                                                                         | R/W |
| b6  | G2INV  | Group 2 Output Polarity Change | 0: Inverted output<br>1: Direct output                                                                                                                                                                         | R/W |
| b7  | G3INV  | Group 3 Output Polarity Change | 0: Inverted output<br>1: Direct output                                                                                                                                                                         | R/W |

- PPG1.PMR

| Bit | Symbol | Bit Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R/W |
|-----|--------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | G0NOV  | Group 4 Non-Overlap | <ul style="list-style-type: none"> <li>When the PPG1.PTRSLR.PTRSL bit is 0               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected MTUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected MTUn)<br/>(n = 0 to 3)</li> </ul> </li> <li>When the PPG1.PTRSLR.PTRSL bit is 1               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected TPUUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected TPUUn)<br/>(n = 0 to 3)</li> </ul> </li> </ul> | R/W |
| b1  | G1NOV  | Group 5 Non-Overlap | <ul style="list-style-type: none"> <li>When the PPG1.PTRSLR.PTRSL bit is 0               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected MTUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected MTUn)<br/>(n = 0 to 3)</li> </ul> </li> <li>When the PPG1.PTRSLR.PTRSL bit is 1               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected TPUUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected TPUUn)<br/>(n = 0 to 3)</li> </ul> </li> </ul> | R/W |
| b2  | G2NOV  | Group 6 Non-Overlap | <ul style="list-style-type: none"> <li>When the PPG1.PTRSLR.PTRSL bit is 0               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected MTUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected MTUn)<br/>(n = 0 to 3)</li> </ul> </li> <li>When the PPG1.PTRSLR.PTRSL bit is 1               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected TPUUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected TPUUn)<br/>(n = 0 to 3)</li> </ul> </li> </ul> | R/W |
| b3  | G3NOV  | Group 7 Non-Overlap | <ul style="list-style-type: none"> <li>When the PPG1.PTRSLR.PTRSL bit is 0               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected MTUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected MTUn)<br/>(n = 0 to 3)</li> </ul> </li> <li>When the PPG1.PTRSLR.PTRSL bit is 1               <ul style="list-style-type: none"> <li>0: Normal operation<br/>(Output values updated on compare match A in the selected TPUUn)</li> <li>1: Non-overlapping operation<br/>(Output values updated on compare match A or B in the selected TPUUn)<br/>(n = 0 to 3)</li> </ul> </li> </ul> | R/W |

| Bit | Symbol | Bit Name                       | Description                            | R/W |
|-----|--------|--------------------------------|----------------------------------------|-----|
| b4  | G0INV  | Group 4 Output Polarity Change | 0: Inverted output<br>1: Direct output | R/W |
| b5  | G1INV  | Group 5 Output Polarity Change | 0: Inverted output<br>1: Direct output | R/W |
| b6  | G2INV  | Group 6 Output Polarity Change | 0: Inverted output<br>1: Direct output | R/W |
| b7  | G3INV  | Group 7 Output Polarity Change | 0: Inverted output<br>1: Direct output | R/W |

PPGn.PMR (n = 0, 1) selects the pulse output mode of the PPG on a group-by-group basis.

While inverted output is selected, a low-level pulse is output when the values in PPGn.PODRH and PPGn.PODRL are 1, and a high-level pulse is output when the values in PPGn.PODRH and PPGn.PODRL are 0.

In addition, when non-overlapping operation is selected, the PPG updates its output values on compare match A or B in an MTU or TPU channel that functions as an output trigger.

For details, refer to section 29.3.4, Pulse Output in Non-Overlapping Operation.

### 29.3 Operation

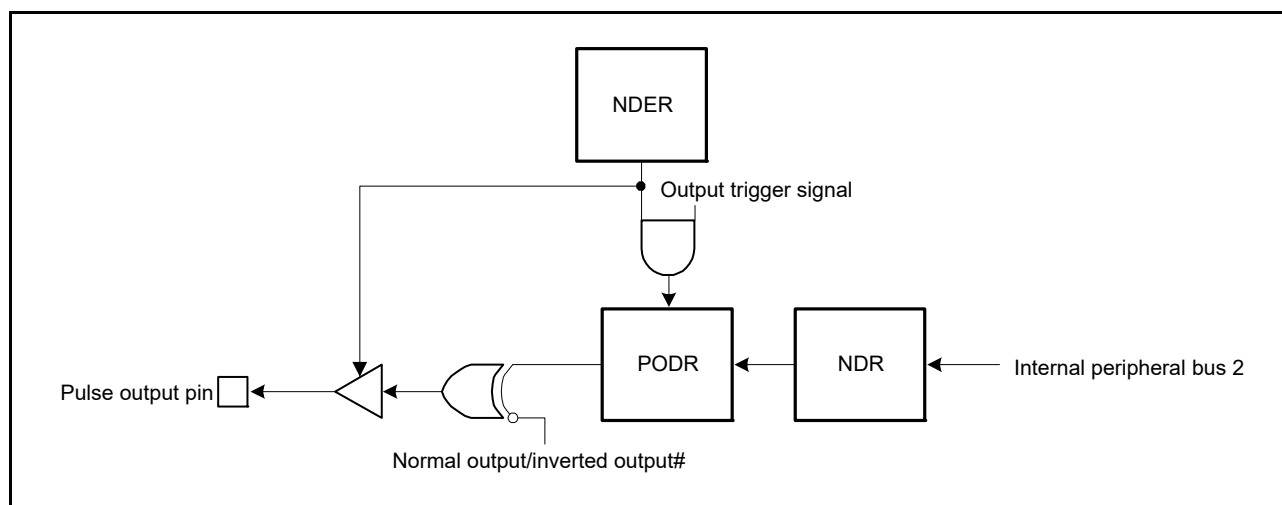
Figure 29.4 shows a schematic diagram of the PPG.

PPG pulse output is enabled when the corresponding bits in PPGn.NDERH and PPGn.NDERL ( $n = 0, 1$ ) are set to 1 (data transfer is enabled).

An initial output value is determined by the initial settings in the corresponding PPGn.PODRH and PPGn.PODRL.

When the compare match event selected in PPGn.PCR occurs, the output values are updated by transfer of the values in the corresponding PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 to PPGn.PODRH and PPGn.PODRL, respectively.

Consecutive output of up to 16 bits of data is possible by writing new output data to PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 before the next compare match.



**Figure 29.4** Schematic Diagram of PPG

### 29.3.1 Output Timing

When the selected compare match event occurs while pulse output is enabled, the values in PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 ( $n = 0, 1$ ) are transferred to PPGn.PODRH and PPGn.PODRL, respectively, and then output on the corresponding pins.

Figure 29.5 shows the timing of the above operation. In this case, the timing when compare match A triggers normal output from groups 2 and 3 is shown.

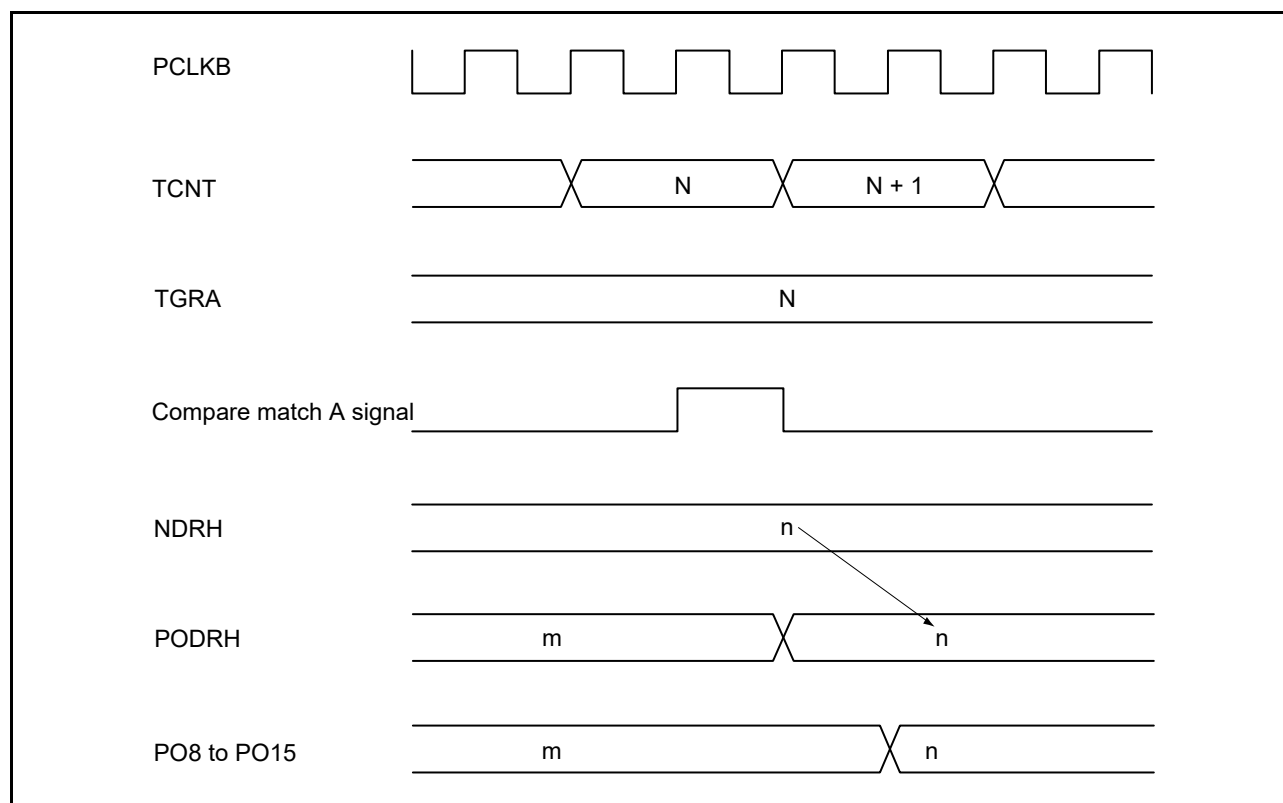


Figure 29.5 Timing of Transfer and Output of the Values in NDR (Example)

### 29.3.2 Example of Setup Procedure for Pulse Output in Normal Operation

Figure 29.6 and Figure 29.7 show examples of setup procedures for pulse output in normal operation.

#### (1) PPG0 Setting

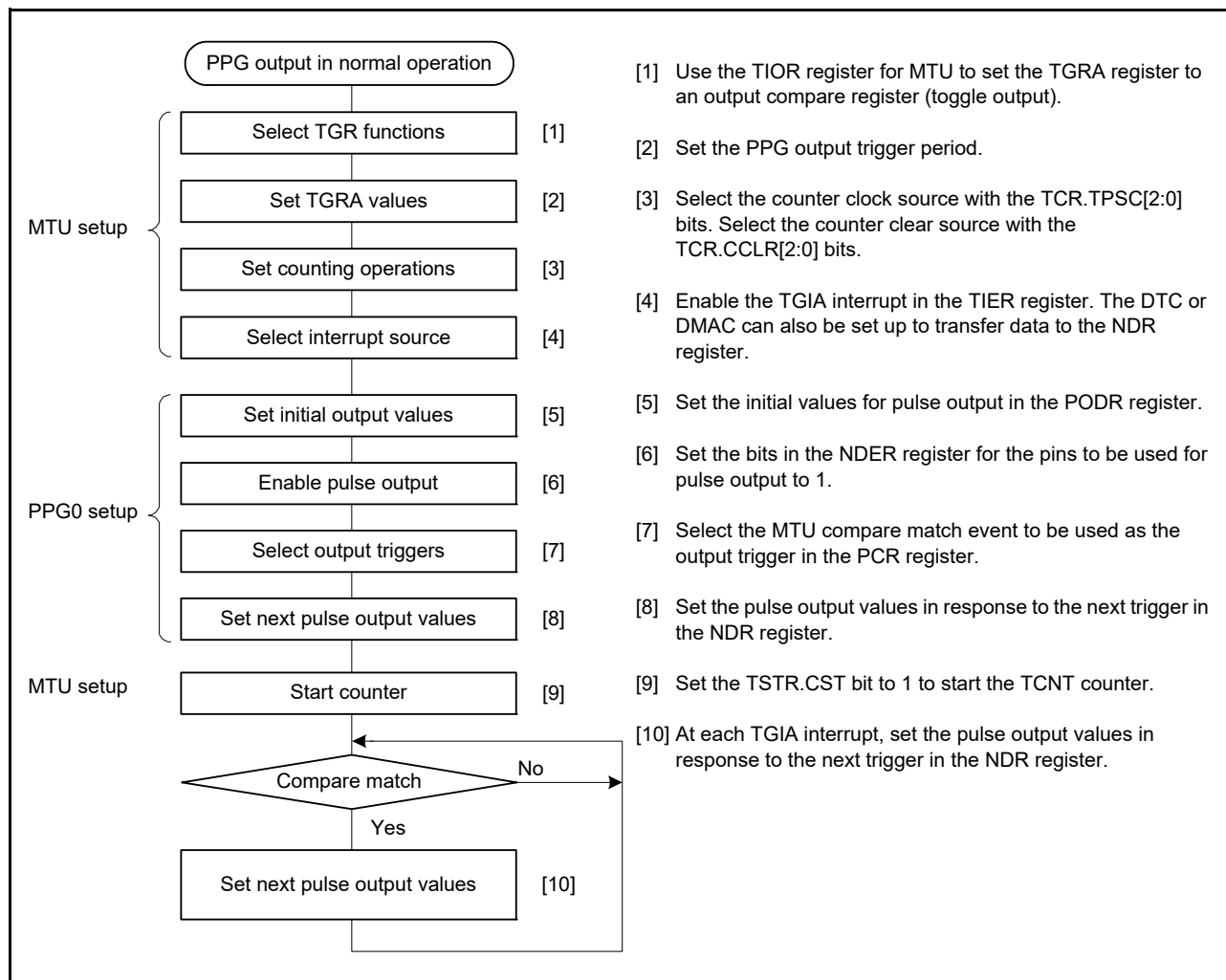


Figure 29.6 Example of Setup Procedure for Pulse Output in Normal Operation (PPG0 Setting)

## (2) PPG1 Setting

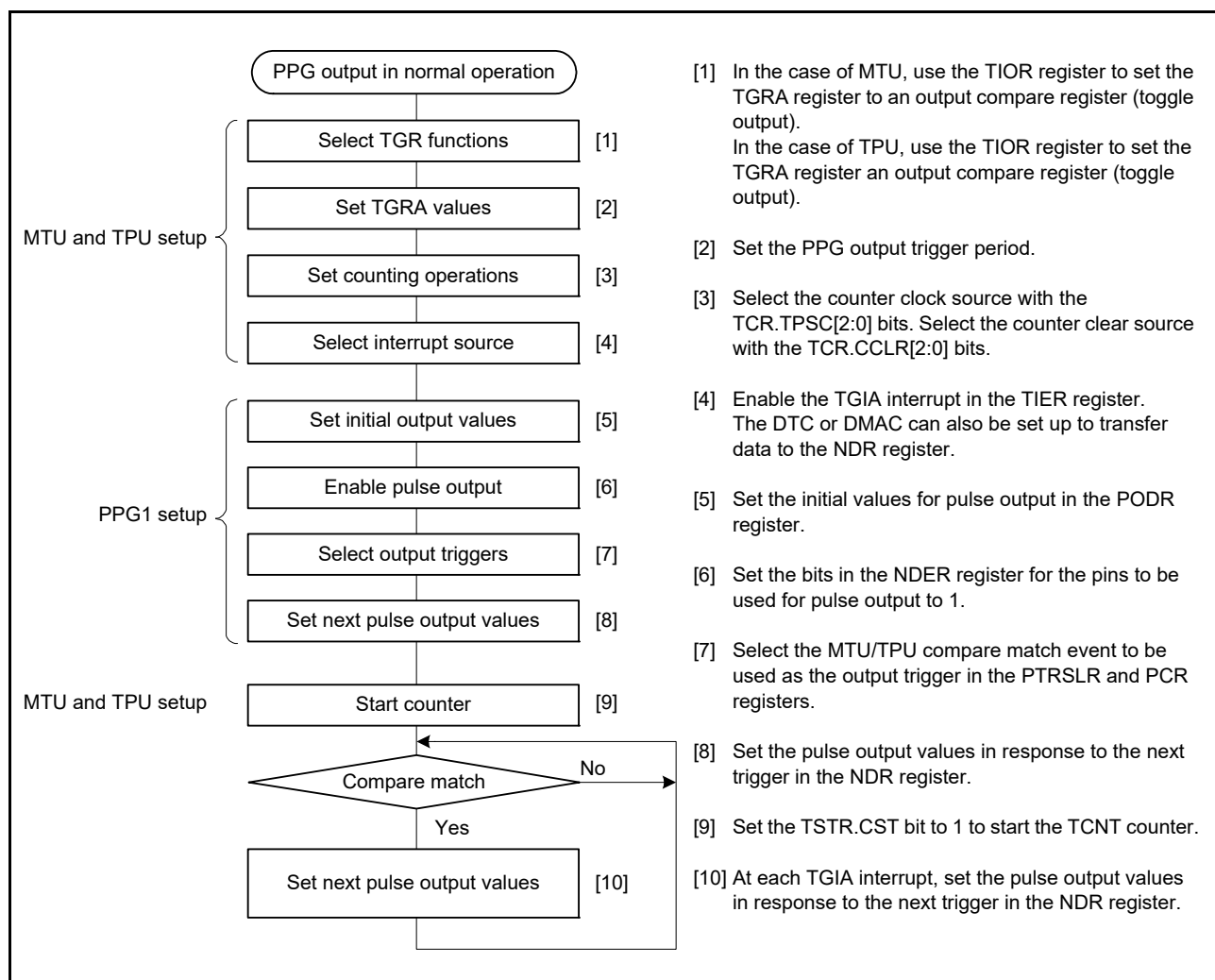
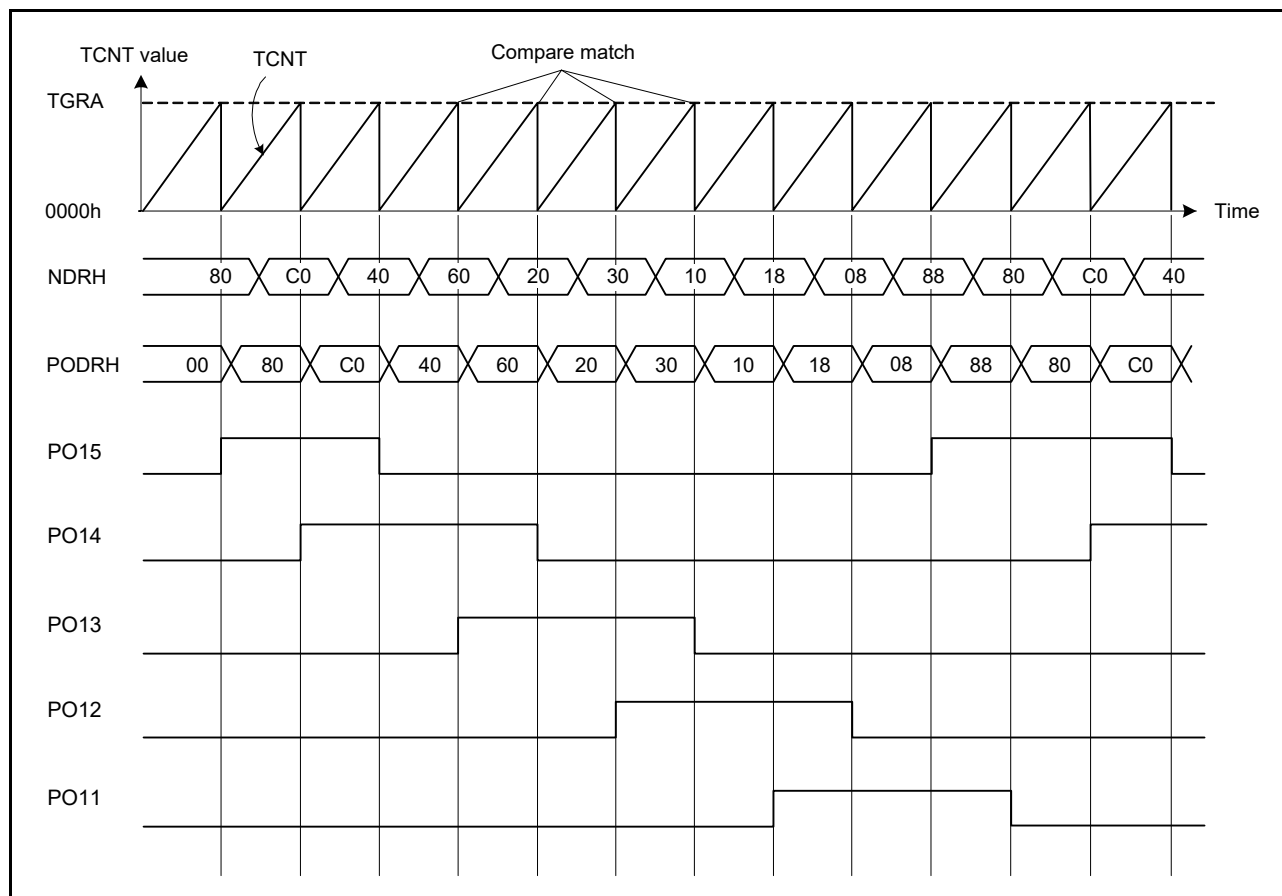


Figure 29.7 Example of Setup Procedure for Pulse Output in Normal Operation (PPG1 Setting)

### 29.3.3 Example of Pulse Output in Normal Operation (Example of Five-Phase Pulse Output)

Figure 29.8 shows an example in which pulse output from the PPG0 is used for cyclic five-phase pulse output.



**Figure 29.8** Example of Pulse Output in Normal Operation (Example of Five-Phase Pulse Output)

1. Set an output compare register of the MTUn.TGRA ( $n = 0$  to 3) so that the corresponding compare match signal is the output trigger. Set a cycle in TGRA so that the counter will be cleared by compare match A. Set the MTUn.TIER.TGIEA bit to 1 to enable the compare match/input capture A (TGIA<sub>n</sub>) interrupt.
2. Write F8h to PPG0.NDRH, and set the G3CMS[1:0] and G2CMS[1:0] bits in PPG0.PCR to select the respective compare matches in the MTUn selected in the previous step to be the output triggers. Write output data 80h to PPG0.NDRH.
3. The timer counter in the MTU starts. When compare match A occurs, the values in PPG0.NDRH are transferred to PPG0.PODRH and output. The TGIA<sub>n</sub> interrupt handling routine writes the next output data C0h to PPG0.NDRH.
4. Five-phase pulse output (one or two phases active at a time) can be obtained subsequently by writing 40h, 60h, 20h, 30h, 10h, 18h, 08h, 88h... at successive TGIA<sub>n</sub> interrupts.

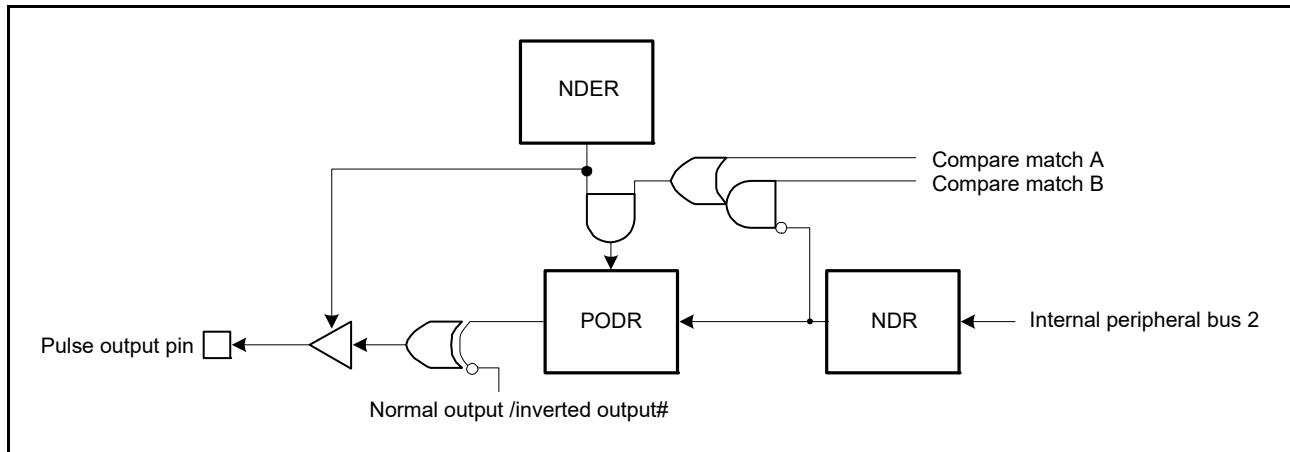
If the DTC or DMAC is set for activation by the TGIA<sub>n</sub> interrupt, pulse output can be obtained without imposing a load on the CPU.

### 29.3.4 Pulse Output in Non-Overlapping Operation

During non-overlapping operation, data transfer from PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 ( $n = 0, 1$ ) to PPGn.PODRH and PPGn.PODRL is performed as follows.

- On compare match A, the values in PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 are transferred to PPGn.PODRH and PPGn.PODRL.
- On compare match B, data transfer proceeds for bits in PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 that have the value 0. It does not proceed for bits having the value 1.

Figure 29.9 shows the pulse output in non-overlapping operation.

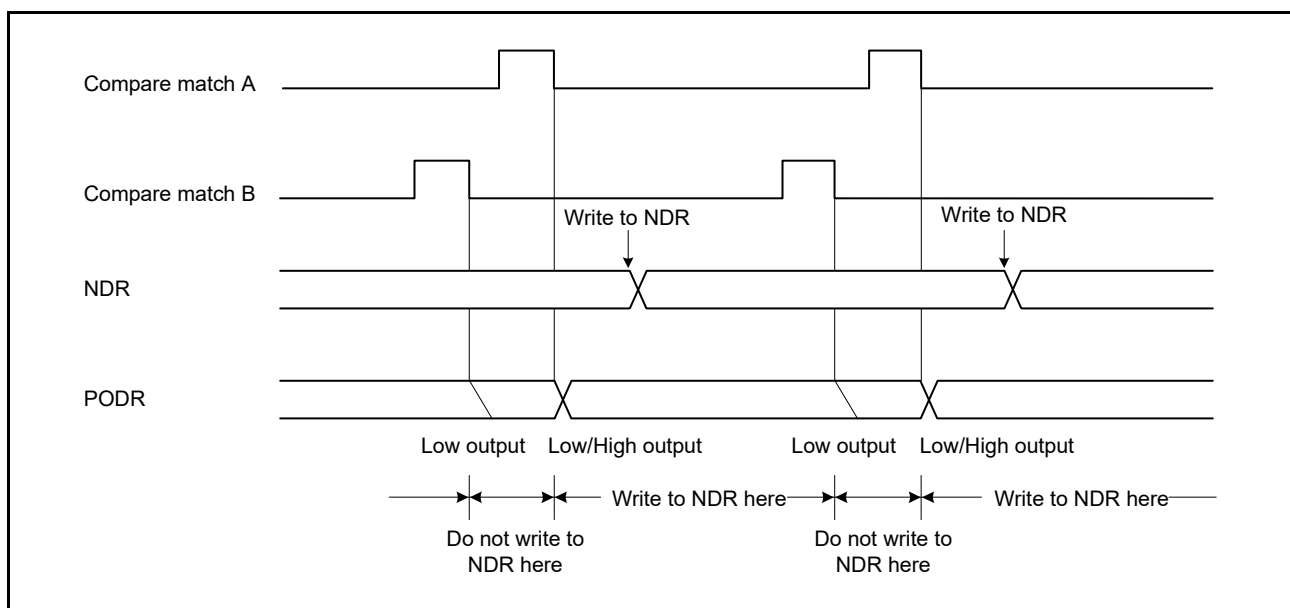


**Figure 29.9** Pulse Output in Non-Overlapping Operation

Therefore, compare match B before compare match A allows 0-valued data to be transferred in advance of 1-valued data. Do not change the values in PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 during the interval from compare match B to compare match A (the non-overlap margin).

To transfer 0-valued data in advance of 1-valued data, write the next data to PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2 from within the TGIA interrupt handling routine or by using a TGIA interrupt to activate transfer by the DTC or DMAC. In any case, the next data must be written before the next compare match B occurs.

Figure 29.10 shows the timing of the above procedure.



**Figure 29.10** Non-Overlapping Operation and Write Timing to PPGn.NDRH, PPGn.NDRL, PPGn.NDRH2, and PPGn.NDRL2

### 29.3.5 Example of Setup Procedure for Pulse Output in Non-Overlapping Operation

Figure 29.11 and Figure 29.12 show examples of setup procedure for pulse output in non-overlapping operation.

#### (1) PPG0 Setting

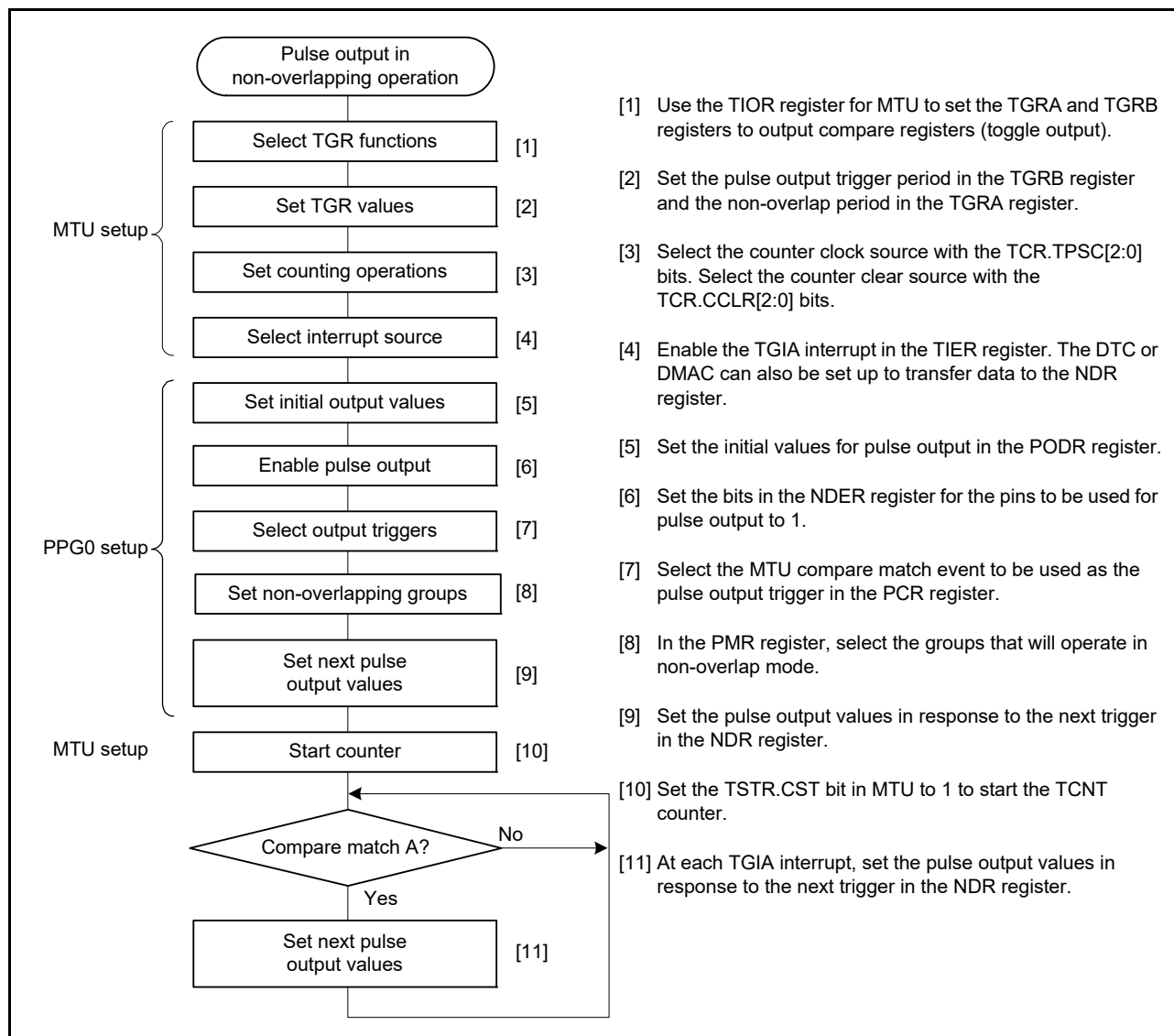


Figure 29.11 Example of Setup Procedure for Pulse Output in Non-Overlapping Operation (PPG0 Setting)

## (2) PPG1 Setting

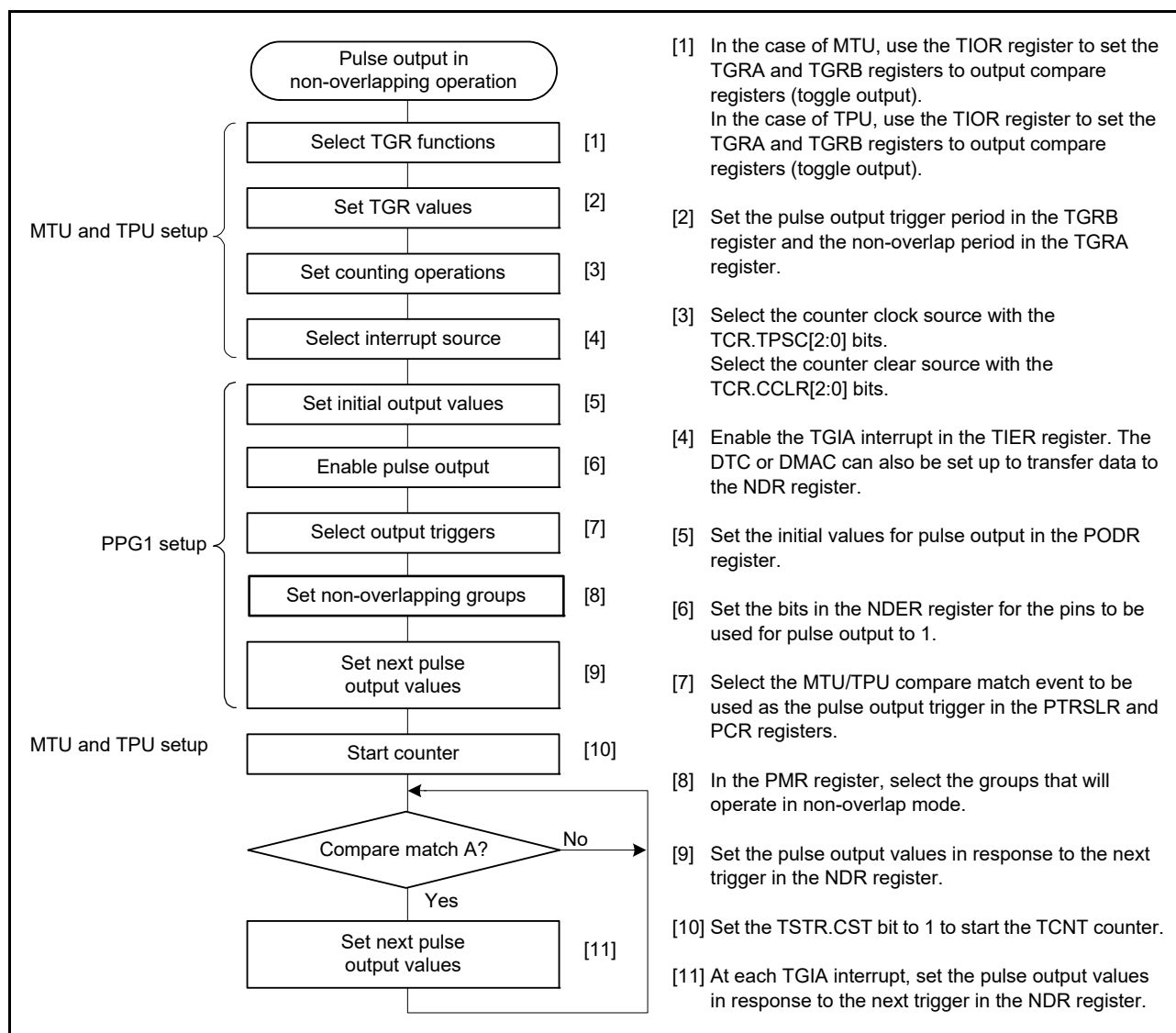
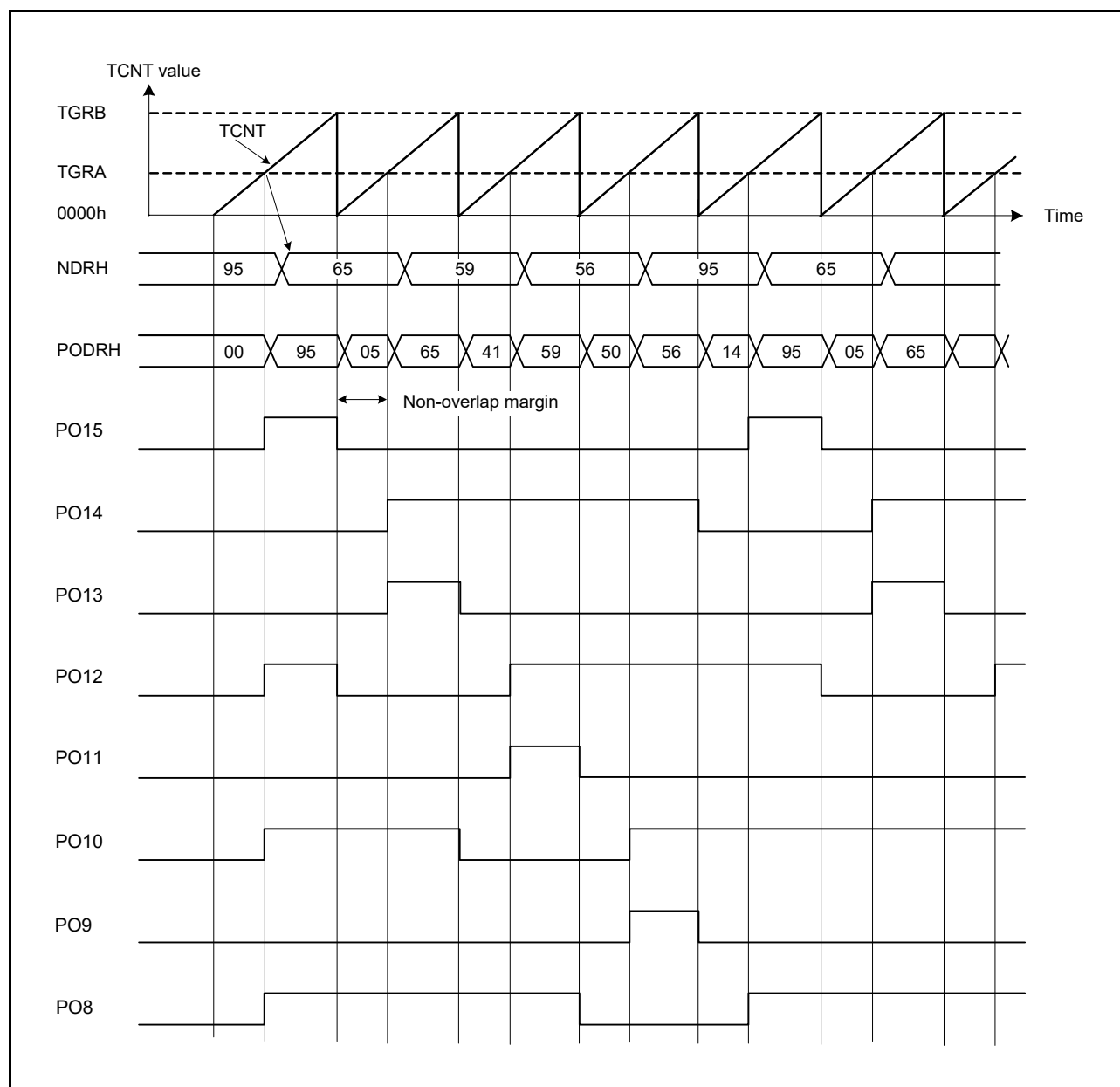


Figure 29.12 Example of Setup Procedure for Pulse Output in Non-Overlapping Operation (PPG1 Setting)

### 29.3.6 Example of Pulse Output in Non-Overlapping Operation (Example of Four-Phase Complementary Non-Overlapping Output)

Figure 29.13 shows an example in which pulse output from the PPG0 is used for four-phase complementary non-overlapping pulse output.



**Figure 29.13** Example of Pulse Output in Non-Overlapping Operation (Four-Phase Complementary Non-Overlapping Output)

1. Set output compare registers of the MTUn.TGRA and MTUn.TGRB ( $n = 0$  to 3) so that the corresponding compare match signals are the output triggers. Set the trigger period in TGRB and the non-overlap margin in TGRA, and set the counter to be cleared by compare match B. Set the MTUn.TIER.TGIEA bit to 1 to enable the compare match/ input capture A (TGIA) interrupt.
2. Write FFh in the PPG0.NDERH register, and set the G3CMS[1:0] and G2CMS[1:0] bits in PPG0.PCR to select the respective compare matches in the MTUn selected in the previous step to be the output triggers.  
Set the G3NOV and G2NOV bits in PPG0.PMR to 1 to select non-overlapping outputs. Write output data 95h in the PPG0.NDRH register.
3. The timer counter in the MTU starts. When a compare match with TGRB occurs, outputs change from high to low. When a compare match with TGRA occurs, outputs change from low to high (the change from low to high is delayed by the value set in TGRA).  
The TGIA interrupt handling routine writes the next output data 65h in the PPG0.NDRH register.
4. Four-phase complementary non-overlapping pulse output can be obtained subsequently by writing 59h, 56h, 95h... at successive TGIA interrupts.  
If the DTC or DMAC is set for activation by the TGIA interrupt, pulse output can be obtained without imposing a load on the CPU.

### 29.3.7 Inverted Pulse Output

When the G3INV, G2INV, G1INV, and G0INV bits in PPG0.PMR are cleared to 0, the values that are the inverse of the respective values in PPG0.PODRH and PPG0.PODRL can be output.

Figure 29.14 shows the outputs when the G3INV and G2INV bits are cleared to 0 in addition to the settings in Figure 29.13.

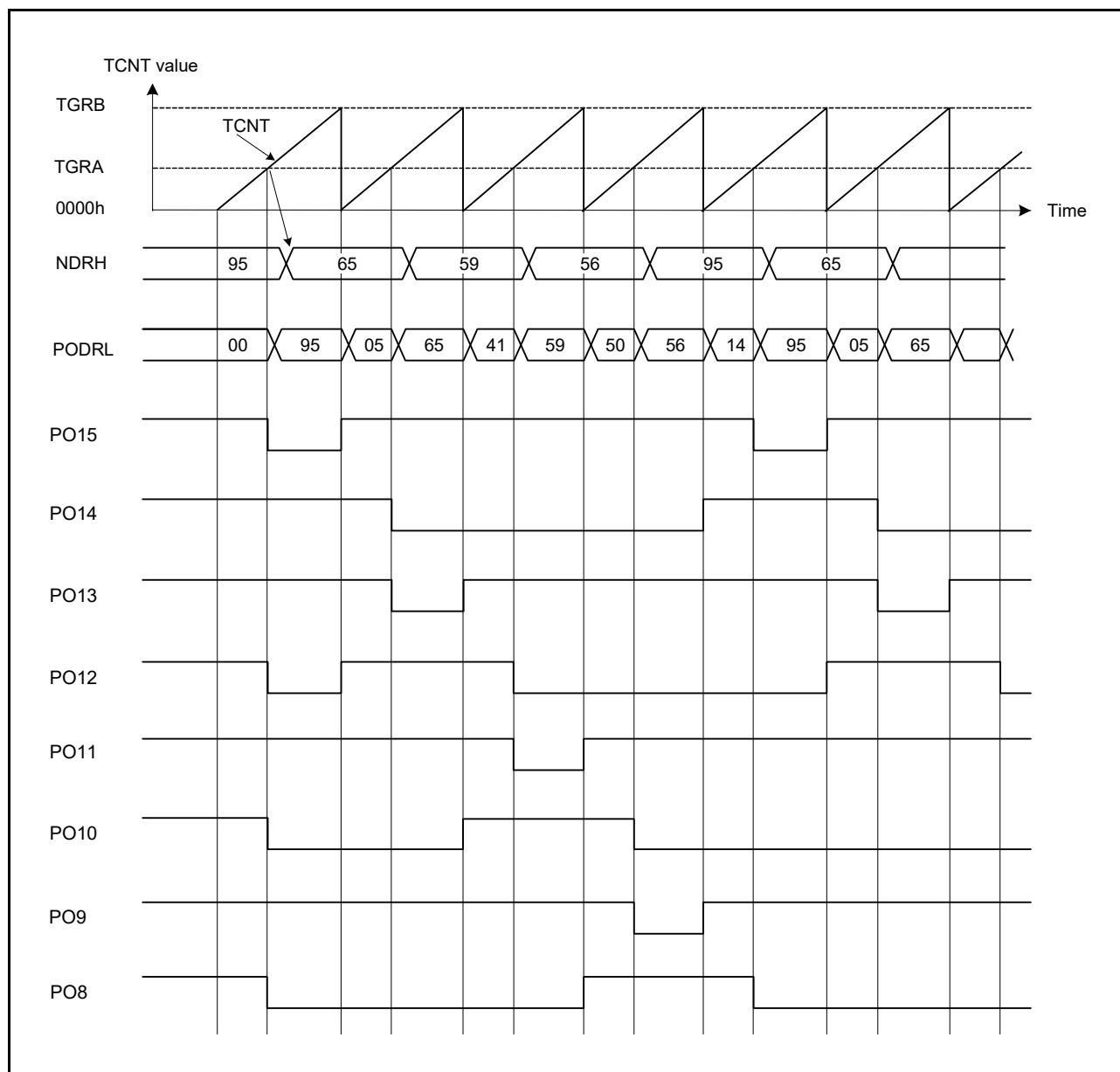


Figure 29.14 Inverted Pulse Output (Example)

### 29.3.8 Pulse Output Triggered by Input Capture

Pulse output from the PPG0 can be triggered by the MTU input capture as well as by compare match. When MTU<sub>n</sub>.TGRA ( $n = 0$  to 3) functions as an input capture register selected by PPG0.PCR, pulse output is triggered by the input capture signal.

Figure 29.15 shows the timing of pulse output triggered by input capture.

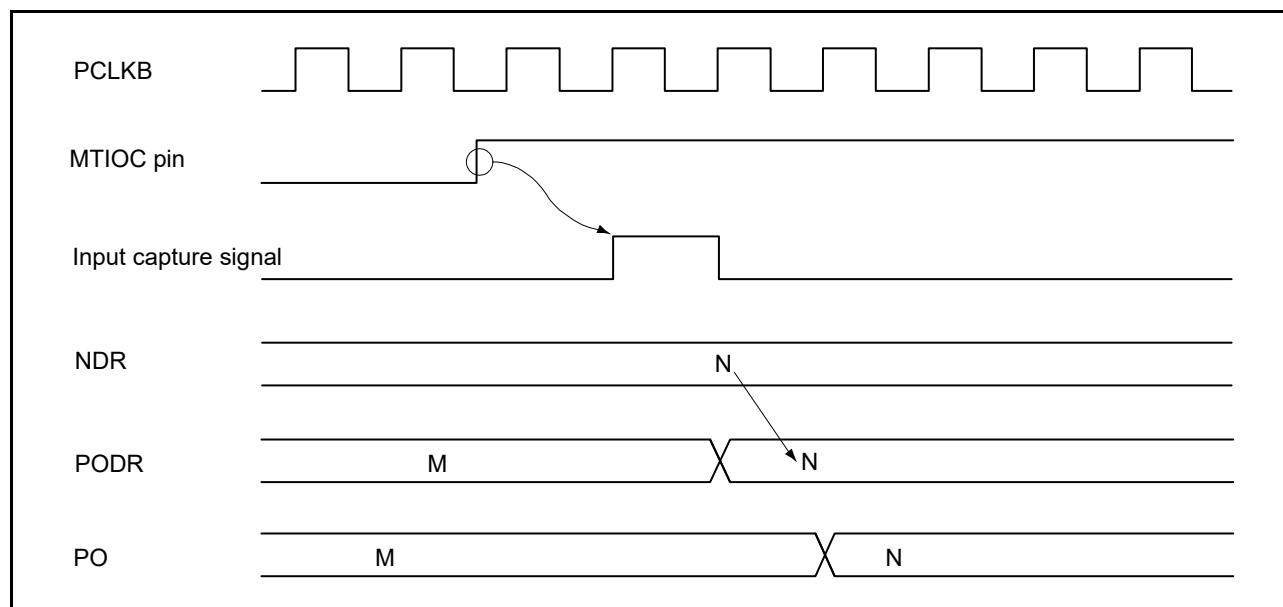


Figure 29.15 Timing of Pulse Output Triggered by Input Capture (Example)

## 29.4 Usage Note

### 29.4.1 Module-Stop Function Setting

Operation of the PPG can be disabled or enabled by the module stop control register. The initial setting is for operation of the PPG to be stopped. Register access is enabled by clearing module-stop state. For details, refer to **section 11, Low Power Consumption**.